Project: Worked examples
Title: Bolted joint (simplified VDI 2230)
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1. Scope and method

Calculates the preload from the tightening torque, the load factor, the residual clamp, separation, yield, thread stripping and fatigue amplitude of one bolt clamping members under an axial working load. It uses a simplified VDI 2230 approach.

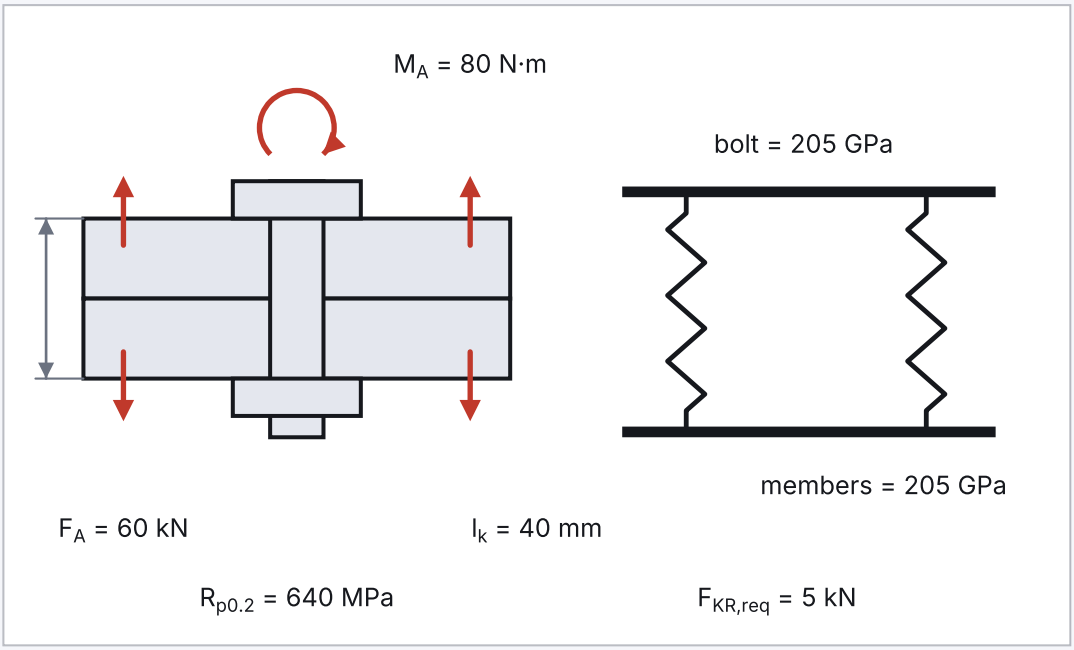


Figure 1 — A preloaded bolt clamping two members, the working load pulling them apart; beside it, bolt and members as two springs in parallel. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Tensile stress area	$A_s = (\pi/4)\cdot((d_2 + d_3)/2)^2$, with $d_2 = d - 0.6495\ p_{thread}$ and $d_3 = d - 1.2269\ p_{thread}$
Assembly preload	$F_M = M_A/(K\ d)$ by the nut factor; $F_M = M_A/(0.16\ p_{thread} + 0.58\ d_2\ \mu_G + 0.5\ D_{Km}\ \mu_K)$ by thread and head friction; $F_M = F_V$ when measured; $F_{M,min} = F_M/\alpha_A$ after the scatter
Thread torsion under preload	$\tau_M = M_G/W_p$ with $M_G = F_M\ (0.16\ p_{thread} + 0.58\ d_2\ \mu_G)$ and $W_p = \pi/16\cdot d_3^3$ at the minor diameter, which is conservative against the VDI 2230 mean diameter d_S
Bolt resilience, the head, free shank and engaged thread as springs in series	$\delta_S = (0.4\ d + 0.4\ d)/(E_S\ A_N) + l_k/(E_S\ A_s) + 0.5\ d/(E_S\ A_{d3})$
Member stiffness, the clamped parts as two pressure frusta	$k = 0.5774\ \pi\ E\ d\ /\ \ln[((1.155\ t + d_w - d_h)(d_w + d_h))/((1.155\ t + d_w + d_h)(d_w - d_h))]$
Load factor and embedding	$\Phi = n_{load}\ k_S/(k_S + k_P)$; $F_Z = f_{embed}\ (k_S\ k_P)/(k_S + k_P)$

WHAT IT CALCULATES	RELATION
Working state	$F_{KR} = \max(F_{M,min} - (1 - \Phi)\, F_A - F_Z,\, 0)$; $F_{S,max} = \max(F_M + \Phi\, F_A,\, F_A)$; $\sigma_{red} = \sqrt{(\sigma_z^2 + 3\,(0.5\,\tau)^2)}$; $\sigma_a = (F_{S,max} - F_M)/(2\,A_s)$
Thread engagement needed against stripping	$m_{min} = S\, F_{S,max}/(0.5\,\pi\, d\, \tau_B)$

Basis: Systematic calculation of highly stressed bolted joints — calculation steps R0–R13[1](§5.1–§5.5); Tensile stress area of the thread[2](§9.1.6); Joint stiffness (frusta); preload and torque[3](§8-5, §8-7, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Preload method	friction		spec: Assembly procedure AP-7	any
d	Nominal thread diameter	12	mm	drawing: DRG-FLG-02 rev A	3 … 64 mm
p_{thread}	Thread pitch	1.75	mm	spec: ISO 261 coarse pitch	0.35 … 6 mm
R_{p02}	Bolt 0.2 % proof (yield) strength	640	MPa	spec: ISO 898-1 class 8.8	200 … 1200 MPa
E_S	Bolt Young's modulus	205	GPa	spec: steel	60 … 250 GPa
$\sigma_{a,limit}$	Bolt endurance limit (stress amplitude)	50	MPa	assumption: Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.	10 … 200 MPa
l_k	Clamp length	40	mm	drawing: DRG-FLG-02 rev A	1 … 2000 mm
d_h	Hole diameter	13	mm	drawing: DRG-FLG-02 rev A	1 … 100 mm
d_w	Head / washer bearing diameter	18	mm	spec: ISO 4762 head bearing	2 … 200 mm
E_P	Clamped members' Young's modulus	205	GPa	spec: steel	5 … 250 GPa
n_{load}	Load introduction factor (0 = at the clamped faces, 1 = under the head)	0.5		assumption: Load introduced at mid-thickness.	0 … 1
f_{embed}	Embedding (settling) loss	11	µm	assumption: Three interfaces, machined.	0 … 100 µm
M_A	Tightening torque	80	N·m	spec: Assembly procedure AP-7	0.1 … 10000 N·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
K	Nut factor (nut-factor method); F_M is the maximum preload, so enter the smallest value the joint can have	0.2		assumption: Not used by the friction method.	0.1 … 0.35
μ_G	Thread friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have	0.12		assumption: Lightly oiled zinc-plated thread.	0.06 … 0.25
μ_K	Head / nut bearing friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have	0.12		assumption: Lightly oiled head bearing.	0.06 … 0.25
ν_{perm}	Permitted assembly utilisation ν: the equivalent stress at assembly over the proof strength (VDI 2230 R7; 0.9 is the usual target, 1.0 is the limit at which the bolt yields in tightening)	1		assumption: Assembly utilisation up to 1.0, where the bolt reaches yield in tightening (VDI 2230 R7 limit); a 0.9 target is usual and can be entered instead.	0.5 … 1
α_A	Tightening scatter factor ($F_{M,max}$ / $F_{M,min}$)	1.6		assumption: Torque wrench, VDI 2230 Table A8 order.	1 … 4
F_A	Axial working load (per bolt)	60	kN	spec: LOAD-CASE-3	0 … 5000 kN
$F_{KR,req}$	Required residual clamp force	5	kN	spec: SPEC-FLG §6 sealing	0 … 5000 kN
τ_B	Nut / internal-thread material shear strength	400	MPa	assumption: Nut material shear strength.	50 … 1200 MPa
m_{eff}	Thread engagement length provided	12	mm	drawing: DRG-FLG-02 rev A	1 … 200 mm
S_{strip}	Safety factor on stripping	1.25		assumption: Stripping safety factor.	1 … 3

3. Assumptions

A1. The joint is concentrically clamped and loaded, and the members behave as two 30° frusta in series. (Eccentric clamping and loading (VDI 2230 full procedure) are outside this sheet.)
A2. Thread torsion is taken at 50 % in the equivalent stress for the working state. (Common VDI 2230 practice (k_τ = 0.5).)
A3. The thread torsion modulus is taken at the minor diameter d_3, which is conservative. (With the VDI 2230 mean diameter the torsion stress is about 14 % lower and ν_M about 5 % lower at the defaults.)
A4. F_M is the maximum preload, from the smallest friction and nut factor, and $F_{M,min}$ is the minimum

after scatter. (VDI 2230 R2: the maximum preload comes from the minimum friction coefficients.)
A5. The working load pulsates from 0 to F_A, so the fatigue amplitude is $\Phi F_A / (2 A_s)$. (Compare it with the bolt's endurance limit as entered.)
A6. Thread stripping uses a shear area of half the pitch per unit length on the nominal diameter. (A conservative simplification of ISO 898-2 / VDI 2230 §5.5.5.)
A7. $\sigma_{a,limit}$ (Bolt endurance limit (stress amplitude)) is taken as 50 MPa.
(Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.)
A8. n_{load} (Load introduction factor (0 = at the clamped faces, 1 = under the head)) is taken as 0.5. (Load introduced at mid-thickness.)
A9. f_{embed} (Embedding (settling) loss) is taken as 11 µm. (Three interfaces, machined.)
A10. K (Nut factor (nut-factor method); F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.2. (Not used by the friction method.)
A11. μ_G (Thread friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled zinc-plated thread.)
A12. μ_K (Head / nut bearing friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled head bearing.)
A13. ν_{perm} (Permitted assembly utilisation ν: the equivalent stress at assembly over the proof strength (VDI 2230 R7; 0.9 is the usual target, 1.0 is the limit at which the bolt yields in tightening)) is taken as 1. (Assembly utilisation up to 1.0, where the bolt reaches yield in tightening (VDI 2230 R7 limit); a 0.9 target is usual and can be entered instead.)
A14. α_A (Tightening scatter factor ($F_{M,max} / F_{M,min}$)) is taken as 1.6. (Torque wrench, VDI 2230 Table A8 order.)
A15. τ_B (Nut / internal-thread material shear strength) is taken as 400 MPa. (Nut material shear strength.)
A16. S_{strip} (Safety factor on stripping) is taken as 1.25. (Stripping safety factor.)

4. Working

[24] $d_2 = d - 0.6495 * p_{thread}$ [2](§9.1.6)
$= 12\ mm - 0.6495 \times 1.75\ mm$
$= 10.863\ mm$
Pitch diameter

[25] $d_3 = d - 1.2269 * p_{thread}$ [2](§9.1.6)
$= 12\ mm - 1.2269 \times 1.75\ mm$
$= 9.8529\ mm$
Minor diameter

[26] $A_s = pi / 4 * ((d_2 + d_3) / 2)^2$ [2](§9.1.6)
$= \pi / 4 \times ((10.863\ mm + 9.8529\ mm) / 2)^2$
$= 84.266\ mm^2$
Tensile stress area

[27] $A_N = pi / 4 * d^2$
$= \pi / 4 \times (12\ mm)^2$
$= 113.1\ mm^2$
Nominal shank area

[28] $A_{d3} = pi / 4 * d_3^2$
$= \pi / 4 \times (9.8529\ mm)^2$
$= 76.247\ mm^2$
Minor-diameter area

[29] $D_{Km} = (d_w + d_h) / 2$
$= (18\ mm + 13\ mm) / 2$
$= 15.5\ mm$
Mean bearing diameter under the head

[30] $F_M = M_A / (0.16 * p_{thread} + 0.58 * d_2 * \mu_G + 0.5 * D_{Km} * \mu_K)$ [1](§5.4.3)
= 80 N·m / (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12 + 0.5 × 15.5 mm × 0.12)
= 40.69 kN
Assembly preload (friction)
[31] $F_{M,min} = F_M / \alpha_A$ [1](R2)
= 40.69 kN / 1.6
= 25.431 kN
Minimum preload after scatter
[32] $\sigma_M = F_M / A_s$
= 40.69 kN / 84.266 mm²
= 482.87 MPa
Preload stress
[33] $M_G = F_M * (0.16 * p_{thread} + 0.58 * d_2 * \mu_G)$ [1](§5.4.3)
= 40.69 kN × (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12)
= 42.158 N·m
Thread torque under preload
[34] $W_p = pi / 16 * d_3^3$
= π / 16 × (9.8529 mm)^3
= 187.81 mm³
Polar section modulus at the minor diameter
[35] $\tau_M = M_G / W_p$
= 42.158 N·m / 187.81 mm³
= 224.47 MPa
Thread torsional stress under preload
[36] $\sigma_{red,M} = sqrt(\sigma_M^2 + 3 * \tau_M^2)$ [1](R8)
= sqrt((482.87 MPa)^2 + 3 × (224.47 MPa)^2)
= 619.94 MPa
Equivalent stress at assembly (full torsion)
[37] $\nu_M = \sigma_{red,M} / R_{p02}$
= 619.94 MPa / 640 MPa
= 0.96866
Utilisation of proof strength at assembly
[38] $\sigma_{perm} = \nu_{perm} * R_{p02}$ [1](R7)
= 1 × 640 MPa
= 640 MPa
Permitted equivalent stress at assembly (permitted ν times the proof strength)
[39] $u_{assembly} = \nu_M / \nu_{perm}$
= 0.96866 / 1
= 0.96866
Assembly utilisation over the permitted ν
[40] $\delta_S = 0.4 * d / (E_S * A_N) + l_k / (E_S * A_s) + 0.5 * d / (E_S * A_{d3}) + 0.4 * d / (E_S * A_N)$ [1](§5.1.1)
= 0.4 × 12 mm / (205 GPa × 113.1 mm²) + 40 mm / (205 GPa × 84.266 mm²) + 0.5 × 12 mm / (205 GPa × 76.247 mm²) + 0.4 × 12 mm / (205 GPa × 113.1 mm²)
= 0.0000031135 m/kN
Bolt resilience (head, shank, engaged thread, nut)
[41] $k_S = 1 / \delta_S$
= 1 / 0.0000031135 m/kN
= 321190 kN/m
Bolt stiffness
[42] $t_f = l_k / 2$
= 40 mm / 2
= 20 mm
Frustum height (two equal frusta)

[43] $k_f = 0.5774 * pi * E_P * d / \ln(((1.155 * t_f + d_w - d_h) * (d_w + d_h)) / ((1.155 * t_f + d_w + d_h) * (d_w - d_h)))$ [3](eq. 8-20)

$= 0.5774 \times \pi \times 205\ \text{GPa} \times 12\ \text{mm} / \ln((1.155 \times 20\ \text{mm} + 18\ \text{mm} - 13\ \text{mm}) \times (18\ \text{mm} + 13\ \text{mm}) / ((1.155 \times 20\ \text{mm} + 18\ \text{mm} + 13\ \text{mm}) \times (18\ \text{mm} - 13\ \text{mm})))$

$= 3815600\ \text{kN/m}$

Stiffness of one frustum

[44] $k_P = k_f / 2$ [3](eq. 8-21)

$= 3815600\ \text{kN/m} / 2$

$= 1907800\ \text{kN/m}$

Clamped members' stiffness (two frusta in series)

[45] $\Phi_k = n_{load} * k_S / (k_S + k_P)$ [1](R3)

$= 0.5 \times 321190\ \text{kN/m} / (321190\ \text{kN/m} + 1907800\ \text{kN/m})$

$= 0.072047$

Load factor

[46] $F_Z = f_{embed} * (k_S * k_P) / (k_S + k_P)$ [1](R4)

$= 11\ \mu\text{m} \times 321190\ \text{kN/m} \times 1907800\ \text{kN/m} / (321190\ \text{kN/m} + 1907800\ \text{kN/m})$

$= 3.024\ \text{kN}$

Preload loss from embedding

[47] $F_{SA} = \Phi_k * F_A$

$= 0.072047 \times 60\ \text{kN}$

$= 4.3228\ \text{kN}$

Additional bolt load from the working load

[48] $F_{PA} = (1 - \Phi_k) * F_A$

$= (1 - 0.072047) \times 60\ \text{kN}$

$= 55.677\ \text{kN}$

Load relieved from the members

[49] $F_{KR} = \max(F_{M,min} - F_{PA} - F_Z, 0\ \text{kN})$ [1](R7)

$= \max(25.431\ \text{kN} - 55.677\ \text{kN} - 3.024\ \text{kN}, 0\ \text{kN})$

$= 0\ \text{kN}$

Residual clamp force (minimum; zero once the joint opens)

[50] $F_{S,max} = \max(F_M + F_{SA}, F_A)$

$= \max(40.69\ \text{kN} + 4.3228\ \text{kN}, 60\ \text{kN})$

$= 60\ \text{kN}$

Maximum bolt load (maximum preload + additional load; the whole working load once the joint opens)

[51] $\sigma_{z,max} = F_{S,max} / A_s$

$= 60\ \text{kN} / 84.266\ \text{mm}^2$

$= 712.03\ \text{MPa}$

Maximum bolt tensile stress

[52] $\sigma_{red,B} = \text{sqrt}(\sigma_{z,max}\text{^}2 + 3 * (0.5 * \tau_M)\text{^}2)$ [1](R8)

$= \text{sqrt}((712.03\ \text{MPa})\text{^}2 + 3 \times (0.5 \times 224.47\ \text{MPa})\text{^}2)$

$= 738.09\ \text{MPa}$

Equivalent stress in the working state (50 % torsion)

[53] $u_{yield} = \sigma_{red,B} / R_{p02}$

$= 738.09\ \text{MPa} / 640\ \text{MPa}$

$= 1.1533$

Yield utilisation in the working state

[54] $\sigma_a = (F_{S,max} - F_M) / (2 * A_s)$ [1](R9)

$= (60\ \text{kN} - 40.69\ \text{kN}) / (2 \times 84.266\ \text{mm}^2)$

$= 114.58\ \text{MPa}$

Fatigue stress amplitude (pulsating 0…F_A load)

[55] $u_{fatigue} = \sigma_a / \sigma_{a,limit}$

$= 114.58\ \text{MPa} / 50\ \text{MPa}$

$= 2.2916$

Fatigue utilisation

[56] $m_{min} = S_{strip} * F_{S,max} / (0.5 * pi * d * \tau_B)$ [1](§5.5.5)
$= 1.25 \times 60\ kN / (0.5 \times \pi \times 12\ mm \times 400\ MPa)$
$= 9.9472\ mm$
Thread engagement needed against stripping
[57] $u_{strip} = m_{min} / m_{eff}$
$= 9.9472\ mm / 12\ mm$
$= 0.82893$
Engagement utilisation
[58] $F_{A,sep} = (F_{M,min} - F_Z) / (1 - \Phi_K)$ [1](R7)
$= (25.431\ kN - 3.024\ kN) / (1 - 0.072047)$
$= 24.147\ kN$
Working load at which the members separate (minimum preload, after embedding)
[59] $F_{proof} = R_{p02} * A_s$
$= 640\ MPa \times 84.266\ mm^2$
$= 53.93\ kN$
Bolt load at the proof (yield) strength
[60] $n_0 = F_{A,sep} / F_A$
$= 24.147\ kN / 60\ kN$
$= 0.40245$
Load factor against separation ($F_{A,sep} / F_A$)
[61] $F_{A,yield} = min((F_{proof} - F_M) / \Phi_K, F_{proof})$
$= min((53.93\ kN - 40.69\ kN) / 0.072047, 53.93\ kN)$
$= 53.93\ kN$
Working load at which the bolt reaches its proof load (separation included)
[62] $n_L = F_{A,yield} / F_A$
$= 53.93\ kN / 60\ kN$
$= 0.89884$
Load factor against axial yield of the bolt, torsion excluded

5. Results

SYMBOL	VALUE	DESCRIPTION
F_M	40.69 kN	Assembly preload (maximum)
F_{KR}	0 kN	Residual clamp
$F_{S,max}$	60 kN	Maximum bolt load
Φ_K	0.072047	Load factor
$\sigma_{red,B}$	738.1 MPa	Working equivalent stress
n_0	0.40244	Load factor against separation
n_L	0.89884	Load factor against axial yield
A_s	84.27 mm²	Tensile stress area
$F_{M,min}$	25.43 kN	Minimum preload
σ_M	482.9 MPa	Preload stress
k_S	321186 kN/m	Bolt stiffness
k_P	1907819 kN/m	Members' stiffness
F_Z	3.024 kN	Embedding loss
F_{SA}	4.323 kN	Additional bolt load
$F_{A,sep}$	24.15 kN	Separation load

SYMBOL	VALUE	DESCRIPTION
$\sigma_{z,max}$	712.0 MPa	Maximum bolt tensile stress
σ_a	114.6 MPa	Fatigue amplitude
m_{min}	9.947 mm	Engagement needed
v_M	0.97	Assembly utilisation of proof strength
$u_{assembly}$	0.97	Assembly utilisation over the permitted v
u_{yield}	1.16	Yield utilisation in the working state
$u_{fatigue}$	2.30	Fatigue utilisation
u_{strip}	0.83	Engagement utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Residual clamp force against separation	0 kN	> 0 kN	—	[1](R7)
FAIL	Required residual clamp force against the residual clamp force	0 kN	≥ 5 kN	—	[1](R7)
FAIL	Fatigue stress amplitude against the endurance limit	114.6 MPa	50 MPa	2.30	[1](R9)
FAIL	Working equivalent stress against the proof strength	738.1 MPa	640 MPa	1.16	[1](R8)
PASS	Assembly equivalent stress against the permitted stress	619.9 MPa	640 MPa	0.97	[1](R7)
PASS	Thread engagement needed against the engagement provided	9.947 mm	12 mm	0.83	[1](§5.5.5)

7. Applicability

WARNING M_A: the preload alone takes the bolt above 90 % of its proof strength. [preload-yield]
WARNING F_{KR}: the residual clamp force is not positive. The joint separates under the working load, and the bolt carries all of it. [separation]

8. Verification

Template mech.bolted-joint@1.3.1 Content hash c143be03cdb54ee7 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Verein Deutscher Ingenieure. Systematic Calculation of Highly Stressed Bolted Joints: Joints With One Cylindrical Bolt. VDI 2230 Part 1:2015. Beuth Verlag; 2015.
2. International Organization for Standardization. Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel—Part 1: Bolts, Screws and Studs With Specified Property Classes—Coarse Thread and Fine Pitch Thread. ISO 898-1:2013. ISO; 2013.
3. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.